

Semantic Boundary Reversion in Generative AI Systems: A Competing-Risks Model of Post-Completion Behaviour and Terminal-State Stability

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28 September 2026

Abstract

Generative AI systems can remain highly task-specific throughout the substantive part of a response and then produce locally unnecessary generic behaviour after the task has effectively been completed. Examples include redundant summaries, unrequested extensions, generic offers of further help, needless clarifications, additional agent actions, or other continuation that adds little or negative task value. We introduce *Semantic Boundary Reversion* (SBR) as a falsifiable behavioural construct for this class of event and develop a mathematical framework that does not assume privileged access to model internals.

The central proposal is to treat semantic task completion as a boundary in a stochastic dynamical process. After an independently estimated completion time τ^* , generation is modelled by competing risks: transition into a generic behavioural mode G versus transition into an absorbing terminal state T . If the cause-specific hazard for generic continuation increases relative to the terminal hazard with token-distance from semantic completion, then SBR is boundary-enriched rather than a position-independent instruction-following failure. This yields direct statistical tests using survival analysis, mixed-effects hazard models, ablation-based context-anchoring measures, and pre-registered behavioural taxonomies.

We further show that terminal stability can be improved by separating semantic completion monitoring from the generative trajectory. Under conditional independence, the probability that both a generator and an independent completion monitor fail to terminate multiplies rather than adds. This elementary result motivates a broader architectural principle: different cognitive functions need not be entrusted to a single autoregressive trajectory. Independent components can preserve orthogonal invariants such as task completion, scope, provenance, and safety while remaining coupled to a generative process. The paper therefore connects a small observable failure mode to a more general theory of heterogeneous, polycentric AI architectures without requiring claims about consciousness or model introspection.

Keywords: generative AI; large language models; dynamical systems; attractor dynamics; termination; agentic AI; semantic completion; competing risks; heterogeneous agents; distributed cognition.

1 Introduction

The end of a generative response is usually treated as an implementation detail: an end-of-sequence token is sampled, a stop string is matched, an external criterion fires, or a token budget is exhausted. In deployed agent systems, termination is often handled by a success signal, an evaluator, a fixed loop bound, or a combination of such conditions. Yet a distinction exists between *syntactic termination* and *semantic completion*. A response can be semantically complete before the model emits an end-of-sequence action, and an agent can satisfy the user’s requested objective before its control loop stops.

This paper focuses on the interval between those two events. The motivating observation is simple: a system may remain strongly conditioned on a task for most of a response and then, after the task is already resolved, emit behaviour that is more generic, less locally relevant, or directly contrary to an explicit request to stop. Such behaviour is easy to dismiss as verbosity or instruction-following noise. The alternative considered here is that the completion boundary is itself a useful dynamical probe.

We introduce the term *Semantic Boundary Reversion* (SBR) for locally unnecessary generic behaviour that appears after substantive semantic work has reached completion. The term is deliberately behavioural. It does not assert that the model contains a literal “generic assistant module”, that reinforcement learning caused the event, or that a particular serving layer is responsible. Those are causal hypotheses to be tested separately.

A dynamical-systems treatment of language models is increasingly plausible as a research methodology. Fernando and Gutthoum analyse transformer residual streams as evolving dynamical systems and report attractor-like, self-correcting trajectories under perturbation [1]. Wang et al. show that repeated paraphrasing can converge to stable attractor cycles [2]. At the multi-agent level, Riedl demonstrates measurable higher-order coordination and synergy among language-model agents [3], while Mitra develops a synchronization-based model for heterogeneous collaborative AI systems [4]. These works do not establish SBR, but they motivate treating generative and multi-agent behaviour in terms of trajectories, basins, perturbations, coupling, and stable regimes rather than only isolated outputs.

Termination itself is also an active technical problem. Standard generation stacks commonly stop on explicit EOS or external stopping criteria, not on a semantic proof that the user’s task is complete. Recent work by Yang et al. shows that disagreement over functionally equivalent EOS actions can produce substantial length inflation in on-policy distillation, demonstrating that learned termination behaviour is itself structurally fragile [5]. SBR addresses a different question: even when syntax and decoding are functioning as intended, does the probability of generic residual behaviour increase specifically after semantic completion?

The contribution of this paper is fourfold. First, it gives SBR an operational definition tied to independently estimated task completion. Second, it formulates a competing-risks model with an explicit cumulative incidence for post-completion reversion. Third, it provides falsifiable predictions and an experimental design capable of distinguishing boundary enrichment from simpler alternatives. Fourth, it derives an architectural implication: terminal-state stability improves when completion monitoring is not perfectly correlated with the generative process, motivating heterogeneous systems that distribute invariants across components.

2 Formal setting

2.1 Task specification and semantic completion

Let a task be represented by a tuple

$$\mathcal{T} = (x, \mathcal{R}, \mathcal{I}), \quad (1)$$

where x is the user or environment input, $\mathcal{R} = \{r_1, \dots, r_m\}$ is a finite set of required semantic conditions, and $\mathcal{I} = \{I_1, \dots, I_k\}$ is a finite set of invariants or hard constraints. Examples of invariants include “do not modify files outside directory D ”, “do not ask a follow-up question”, “preserve the numerical result”, or “do not take an external action without approval”.

Let $y_{1:t}$ denote the generated response prefix at token step t . For each semantic requirement, let

$$q_i(y_{1:t}, x) \in [0, 1] \quad (2)$$

be a completion score produced by a pre-registered human annotation protocol or an independent evaluator. Hard invariants are Boolean predicates

$$I_j(y_{1:t}, x) \in \{0, 1\}. \quad (3)$$

Definition 1 (Semantic completion time). *For thresholds $\epsilon \in (0, 1)$ and $\delta \geq 0$, define the semantic completion time as*

$$\tau^* = \inf \left\{ t : \min_i q_i(y_{1:t}, x) \geq 1 - \epsilon, \prod_j I_j(y_{1:t}, x) = 1, \Delta U_t \leq \delta \right\}, \quad (4)$$

where ΔU_t is the expected marginal task utility of continuing generation under the evaluation protocol.

The purpose of ΔU_t is to distinguish a merely sufficient prefix from one for which further generation is no longer expected to improve the requested outcome. In experiments, τ^* should be estimated without access to the model’s hidden chain of thought. A practical protocol can ask multiple blinded annotators to identify the earliest prefix that fully satisfies the explicit task while respecting scope and style constraints.

2.2 Semantic Boundary Reversion

Let B_t be a behavioural class assigned to the generated segment after time t . We distinguish three coarse modes:

$$L : \text{locally task-specific continuation}, \quad (5)$$

$$G : \text{generic or task-extraneous continuation}, \quad (6)$$

$$T : \text{terminal/no further semantic action}. \quad (7)$$

Definition 2 (Semantic Boundary Reversion). *An SBR event occurs if, for some $t > \tau^*$, the system produces an action or segment classified as G before entering T , and the segment has non-positive marginal task utility under the pre-registered evaluation criterion.*

The definition intentionally excludes useful post-completion additions. A concise caveat that prevents a material error may still have positive utility. Conversely, a grammatically polite continuation can count as SBR if it violates the task’s stopping constraint or unnecessarily reopens a completed interaction.

2.3 Terminal invariants

A completed task should ideally enter an absorbing semantic state. Let $\mathcal{A}_T$ denote the set of states in which no further task-relevant action is required.

Definition 3 (Terminal invariant). *A terminal invariant is satisfied when, after τ^* , every subsequent system action is either a null action, a syntactic closure action, or an explicitly requested post-completion operation. Any unrequested semantic expansion is a terminal-invariant violation.*

This definition generalises beyond chat. In a coding agent, a terminal violation might be an unrequested refactor after the requested patch and tests are complete. In a scheduling agent it could be another calendar action after the requested booking has been confirmed.

3 A competing-risks model of the completion boundary

We now formulate the minimal mathematical model needed to make SBR testable. The model is deliberately coarse-grained. It describes observable event dynamics and should not be interpreted as a literal map of internal neural states.

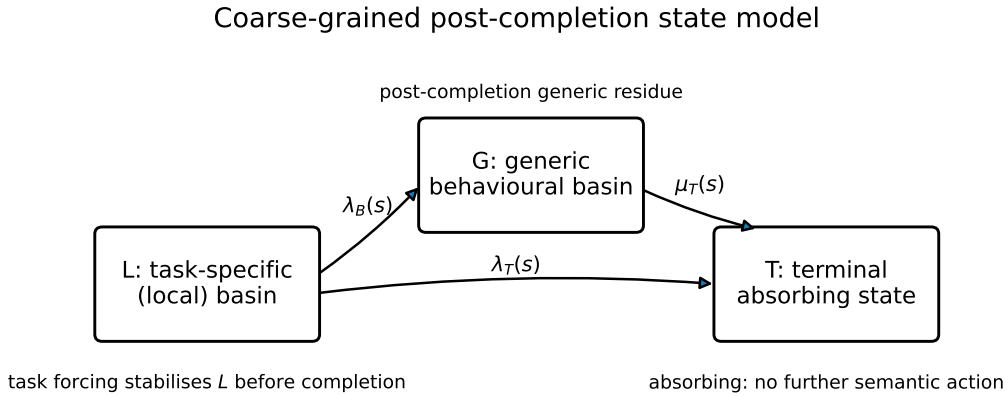


Figure 1: Coarse-grained post-completion model. The task-specific mode L can transition either to generic continuation G at cause-specific hazard $\lambda_B(s)$ or to terminal state T at hazard $\lambda_T(s)$. Generic continuation may itself terminate at rate $\mu_T(s)$. The state T is absorbing.

Let

$$s = t - \tau^* \geq 0 \quad (8)$$

be token-distance, clause-distance, or normalised generation-distance after semantic completion. Consider two competing event times:

$$T_B : \text{time from completion to first SBR event,} \quad (9)$$

$$T_T : \text{time from completion to terminal absorption.} \quad (10)$$

Let $\lambda_B(s)$ and $\lambda_T(s)$ be their cause-specific hazards. Define the event-free survival function

$$S(s) = \exp \left[- \int_0^s (\lambda_B(u) + \lambda_T(u)) du \right]. \quad (11)$$

Proposition 1 (Cumulative incidence of SBR). *The probability that an SBR event has occurred by distance s before terminal absorption is*

$$F_B(s) = \int_0^s S(u) \lambda_B(u) du. \quad (12)$$

The eventual probability of SBR before termination is

$$P(T_B < T_T) = \int_0^\infty S(u) \lambda_B(u) du, \quad (13)$$

provided the total integrated hazard diverges so that an event occurs almost surely.

Proof. For a competing-risks process, $S(u)$ is the probability that neither event has occurred before u . Multiplying by the instantaneous cause-specific hazard $\lambda_B(u)du$ gives the probability of first observing SBR in $[u, u + du)$. Integration gives the cumulative incidence. The infinite-horizon expression follows by taking $s \rightarrow \infty$. $\square$

Corollary 1 (Constant-hazard special case). *If $\lambda_B(s) = a$ and $\lambda_T(s) = c$ for constants $a, c > 0$, then*

$$P(T_B < T_T) = \frac{a}{a + c}. \quad (14)$$

This elementary ratio clarifies the architecture problem: SBR can be reduced either by lowering the reversion hazard a or by increasing the semantic terminal hazard c .

3.1 Boundary enrichment

The central empirical hypothesis of this paper is not merely that SBR occurs, but that it becomes more likely as the trajectory moves away from semantic completion without terminating.

Define the instantaneous conditional probability that the *next* event is SBR rather than termination:

$$\pi_B(s) = \frac{\lambda_B(s)}{\lambda_B(s) + \lambda_T(s)}. \quad (15)$$

Let the hazard ratio be

$$r(s) = \frac{\lambda_B(s)}{\lambda_T(s)}. \quad (16)$$

Then $\pi_B(s) = r(s)/(1 + r(s))$.

Proposition 2 (Monotone boundary enrichment). *If $r(s)$ is differentiable and $r'(s) > 0$ on an interval after semantic completion, then $\pi_B(s)$ is strictly increasing on that interval.*

Proof. Differentiating gives

$$\pi'_B(s) = \frac{r'(s)}{(1 + r(s))^2}. \quad (17)$$

The denominator is positive. Therefore $r'(s) > 0$ implies $\pi'_B(s) > 0$. $\square$

A convenient parametric alternative is

$$\lambda_B(s) = \lambda_{B0} \exp(\beta s), \quad \lambda_T(s) = \lambda_{T0}, \quad (18)$$

for which $\beta > 0$ encodes boundary enrichment and $\beta = 0$ is a position-independent null.

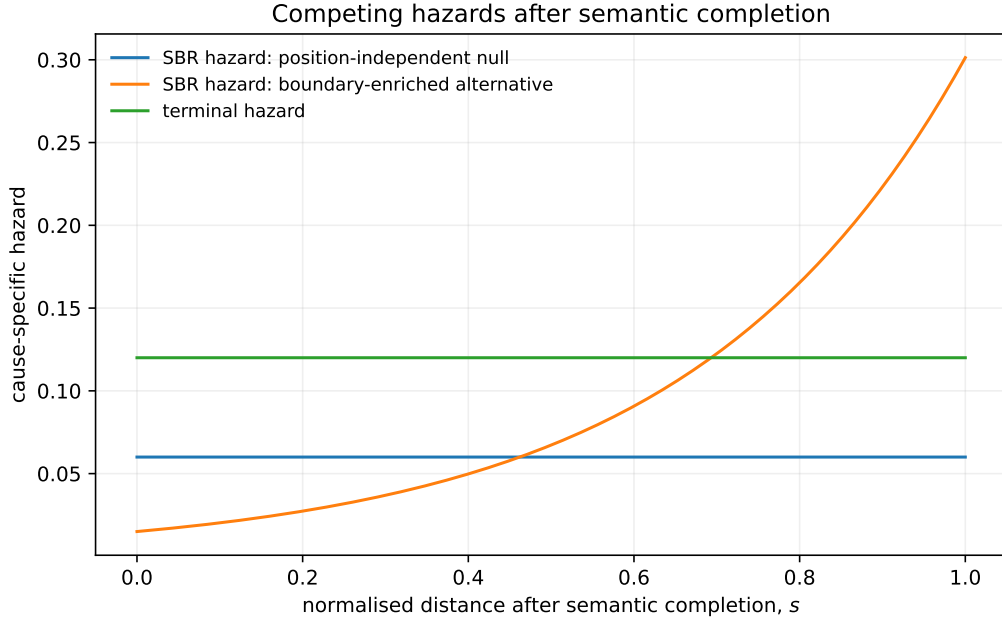


Figure 2: Illustrative synthetic hazards. Under the null, generic residual behaviour has no special relationship to post-completion distance. Under the SBR hypothesis, its cause-specific hazard rises relative to the terminal hazard. These curves are conceptual, not empirical results.

4 A coarse-grained attractor interpretation

The competing-risks model is sufficient for measurement, but an attractor interpretation provides a useful bridge to broader work on model dynamics. Let $z_t \in \mathcal{Z}$ be a coarse-grained system state and let $m_t \in \{L, G, T\}$ denote a behavioural regime. Consider a switched stochastic dynamical system

$$z_{t+1} = F_{m_t}(z_t, u_t) + \xi_t, \quad (19)$$

where u_t represents task-specific forcing from context, explicit constraints, tool state, and unresolved semantic requirements; ξ_t captures stochasticity and omitted variables.

Before completion, unresolved task content can maintain a strong forcing term u_t that keeps the trajectory inside a task-specific basin. At τ^* , the semantic residual drops sharply. If no explicit terminal mechanism absorbs the process, the relative influence of broader learned regularities can increase. In this effective description, SBR corresponds to an escape from a locally conditioned basin $\mathcal{B}_L$ into a broader basin $\mathcal{B}_G$ before absorption in $\mathcal{A}_T$.

This interpretation is intentionally agnostic about mechanism. The generic basin could reflect pre-training statistics, post-training preferences, instruction hierarchy, decoding, system prompts, serving-layer conventions, or an interaction among them. A language model cannot infer which of these caused a particular event merely by generating an introspective explanation. The observable claim is only that a trajectory changes behavioural regime in a way that may be position-dependent.

Attractor-like descriptions of model computation are increasingly used in empirical work. Residual-stream perturbation studies report self-correcting trajectories and attractor-like behaviour [1]; iterative paraphrasing can settle into stable cycles [2]. The present framework extends the same language to the temporal structure of a single response, but treats it as a falsifiable effective model rather than as proof of a particular internal mechanism.

5 Observable quantities and estimators

5.1 Boundary enrichment ratio

For a fixed post-completion window W_+ and a matched pre-completion window W_- , define

$$\text{BER} = \frac{P(B_t = G \mid t \in W_+)}{P(B_t = G \mid t \in W_-)}. \quad (20)$$

A value greater than one is suggestive but not sufficient, because pre- and post-completion text differ semantically. The primary analysis should therefore use hazard regression conditioned on task, model, length, temperature, and instruction regime.

A discrete-time cause-specific hazard can be fitted as

$$\text{logit } P(B_t = G \mid \text{event-free to } t) = \alpha + \beta s_t + \boldsymbol{\gamma}^\top \mathbf{x}_t + b_{\text{task}} + b_{\text{model}}, \quad (21)$$

where β is the boundary-enrichment coefficient and the random effects absorb repeated-task and repeated-model dependence.

The principal hypothesis is

$$H_0 : \beta = 0, \quad H_1 : \beta > 0. \quad (22)$$

5.2 Context anchoring

For open models that expose next-token distributions, a useful measure of local task anchoring is the divergence between the full-context next-token distribution and an ablated-context distribution. Let

$$p_t^{\text{full}} = p(y_t \mid x, y_{<t}), \quad (23)$$

$$p_t^{\text{abl}} = p(y_t \mid x_{\text{abl}}, y_{<t}), \quad (24)$$

where x_{abl} removes task-specific or personalization-specific information while preserving generic conversational form. Define

$$A_t = D_{\text{JS}}(p_t^{\text{full}} \parallel p_t^{\text{abl}}), \quad (25)$$

using Jensen–Shannon divergence.

The SBR hypothesis predicts that, for events genuinely associated with loss of local conditioning, A_t should fall before or during reversion more strongly than in matched non-SBR controls. This is not required for the behavioural definition; it is an optional mechanistic observable.

5.3 Terminal-invariant violation rate

Across N completed tasks, define

$$\text{TIVR} = \frac{1}{N} \sum_{n=1}^N \mathbf{1}\{\text{unrequested semantic action after } \tau_n^*\}. \quad (26)$$

For agentic systems, the indicator should include tool calls and external actions, not only natural-language tokens.

6 Independent termination as an orthogonal stabiliser

The most direct architectural implication of SBR is that semantic completion should not necessarily be entrusted to the same process that generates the content. Let C_G be the event that the generator continues unnecessarily after semantic completion, with probability p_G . Let C_E be the event that an independent semantic completion monitor fails to stop that continuation, with probability p_E .

Suppose the system terminates when either the generator terminates naturally or the monitor asserts semantic completion. Residual continuation then requires both failures.

Proposition 3 (Multiplicative residual risk under conditional independence). *If C_G and C_E are conditionally independent given the completed task state, then*

$$P(C_G \cap C_E) = p_G p_E. \quad (27)$$

For $0 < p_E < 1$, this is strictly smaller than p_G .

Proof. Conditional independence gives $P(C_G \cap C_E) = P(C_G)P(C_E) = p_G p_E$. Since $p_E < 1$, $p_G p_E < p_G$ whenever $p_G > 0$. $\square$

For correlated errors,

$$P(C_G \cap C_E) = p_G p_E + \rho \sqrt{p_G(1-p_G)p_E(1-p_E)}, \quad (28)$$

where ρ is the Bernoulli correlation when the joint distribution is feasible. This expression makes an architectural point explicit: adding another component is most useful when it fails differently. Merely cloning the same generator can leave failure correlation high.

For n conditionally independent monitors with false-negative probabilities $p_1, \dots, p_n$, residual continuation scales as

$$P(\text{all fail}) = p_G \prod_{i=1}^n p_i, \quad (29)$$

although premature termination risk generally increases if the monitors have false positives. Thus the objective is not maximum redundancy but *diverse, calibrated, partially independent judgement*.

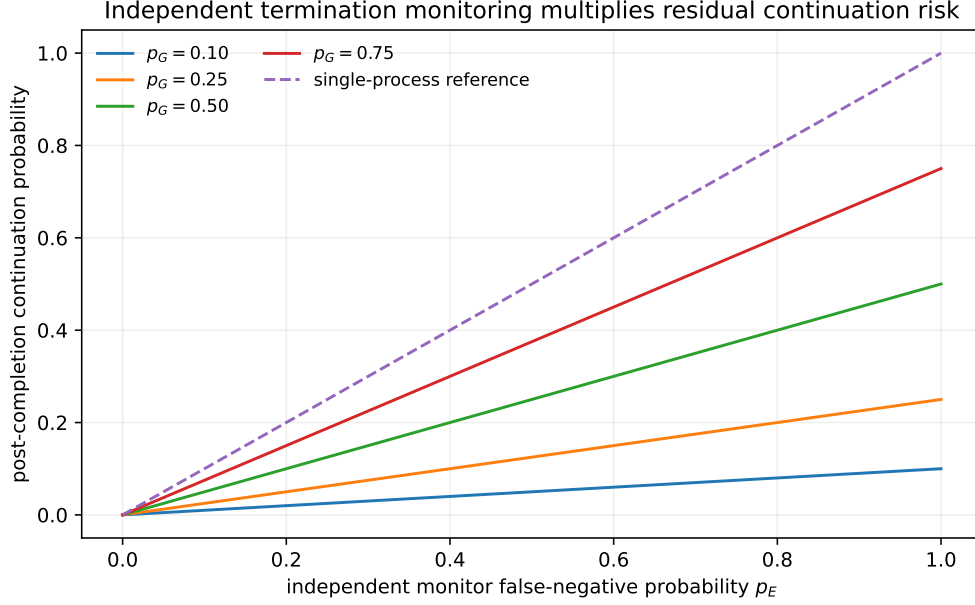


Figure 3: Under conditional independence, residual post-completion continuation is $p_G p_E$. The figure shows how an independent completion monitor can reduce continuation risk for several generator continuation probabilities p_G . Independence is an idealisation; correlated failures reduce the benefit.

The same idea can be represented within the competing-risks model by adding a monitor-induced terminal hazard $\lambda_E(s)$:

$$\lambda_T^{\text{aug}}(s) = \lambda_T(s) + \lambda_E(s). \quad (30)$$

The corresponding SBR cumulative incidence becomes

$$F_B^{\text{aug}}(s) = \int_0^s \exp \left[- \int_0^u (\lambda_B(v) + \lambda_T(v) + \lambda_E(v)) dv \right] \lambda_B(u) du. \quad (31)$$

Whenever $\lambda_E(s) > 0$, the event-free survival available to SBR is reduced.

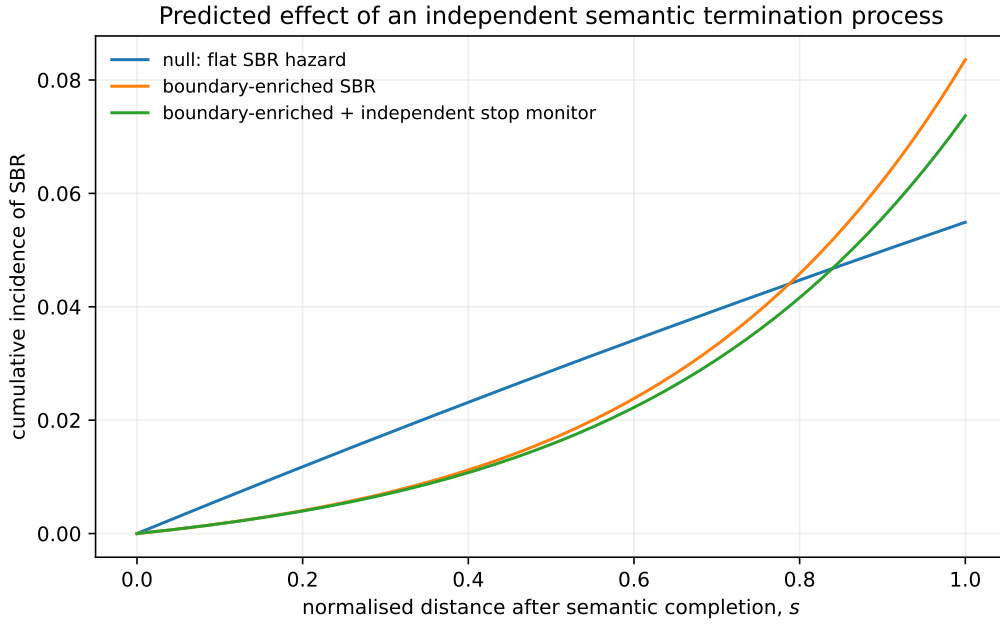


Figure 4: Illustrative synthetic cumulative incidence. A boundary-enriched SBR process produces greater post-completion reversion than a flat-hazard null. Adding an independent semantic termination hazard reduces the cumulative incidence. These are model illustrations, not measured data.

7 Experimental protocol

A useful SBR experiment should be designed to falsify the boundary hypothesis rather than merely collect amusing examples. We propose the following protocol.

7.1 Task construction

Create a stratified task set with explicit natural completion points. Suggested strata include factual explanation, rewriting, coding, planning, document transformation, tool-using agent tasks, and multi-turn personalized tasks. Each task should contain a machine-readable requirement set $\mathcal{R}$ and invariant set $\mathcal{I}$. At least one condition should explicitly instruct the model not to ask follow-up questions or add optional work after completion; another should omit that instruction.

7.2 Completion annotation

For each generated response, blinded annotators should inspect prefixes and identify τ^* . Inter-rater agreement should be reported. A secondary model-based judge may be used for scale but should not be the sole estimator if it shares architecture or training lineage with the tested model.

7.3 Behaviour taxonomy

The post-completion taxonomy should be pre-registered. Candidate categories include:

1. unnecessary follow-up or offer;
2. redundant summary or conclusion restatement;
3. unrequested scope expansion;
4. generic safety or caution appendix not justified by task risk;
5. unsolicited clarification request after sufficient information was already available;
6. unrequested external action or tool call;
7. harmless syntactic closure;
8. useful exception or correction.

Only categories with non-positive marginal task utility should count as SBR.

7.4 Interventions

A factorial design should vary:

- model family and size;
- decoding temperature and nucleus sampling;
- context length and personalization depth;
- explicit “stop when complete” instructions;
- response-length targets;
- semantic completion monitor present/absent;
- homogeneous versus architecturally distinct monitor;
- single-agent versus tool-using agent loop.

The key interaction is between post-completion distance and experimental condition. If SBR is genuinely boundary-enriched, the slope in s should remain after controlling for total response length and should attenuate when an independent terminal stabiliser is added.

7.5 Null hypotheses

Several alternatives should be tested explicitly:

$$H_{\emptyset,1} : \text{generic behaviour is position-independent,} \quad (32)$$

$$H_{\emptyset,2} : \text{apparent enrichment is caused only by response length,} \quad (33)$$

$$H_{\emptyset,3} : \text{annotation of } \tau^* \text{ creates the effect,} \quad (34)$$

$$H_{\emptyset,4} : \text{SBR is ordinary instruction-following noise,} \quad (35)$$

$$H_{\emptyset,5} : \text{only decoding or EOS configuration matters.} \quad (36)$$

The recent EOS-mismatch results of Yang et al. [5] make the final null especially important. A valid SBR claim should survive conditions in which syntactic termination is correctly configured.

8 From a local failure mode to heterogeneous cognitive architecture

SBR begins as a small behavioural observation, but its architectural implication is broader. A monolithic generative process is frequently asked to perform several logically distinct roles simultaneously: solve the task, remember user constraints, preserve scope, decide whether more work is needed, monitor safety, maintain provenance, and decide when to stop. If these functions share highly correlated failure modes, additional scale may improve capability without proportionally improving control.

The mathematical result in Section 6 suggests a different design principle: assign orthogonal invariants to partially independent components and couple them through explicit interfaces. A generative component may optimize semantic construction, a completion monitor may estimate whether the requested state has been achieved, a scope monitor may reject unrequested expansion, and a provenance component may preserve source attribution. Their outputs need not be averaged into a single vote; they can have different authority over different invariants.

This architecture is not equivalent to a committee of identical agents. Diversity matters because the reduction in joint failure depends on imperfect correlation. The same principle appears at larger scales in recent multi-agent work. Riedl’s information-theoretic analysis distinguishes mere temporal coupling from performance-relevant synergy and finds that differentiated identities can support complementary group behaviour [3]. Mitra’s synchronization model likewise treats heterogeneous capabilities and coupling structure as first-class variables rather than assuming identical agents [4].

The resulting research direction is a polycentric cognitive architecture: several specialised processes remain distinct, preserve different invariants, and coordinate without collapsing all control into one generative trajectory. Such systems create new questions about emergent attractors, cross-component entrainment, provenance, veto rights, and collective termination. SBR is therefore useful not because it is a severe failure in isolation, but because it provides a compact experimental probe of a more general issue: *how stable is a locally conditioned cognitive regime when the semantic force that created it has disappeared?*

9 Relation to attractor and collective-dynamics literature

Three bodies of work are especially relevant. First, model-internal dynamical studies provide evidence that transformer representations can be analysed as trajectories with structured and partially self-correcting geometry

[1]. Second, iterative language transformations can produce stable text-space attractor cycles [2]. Third, multi-agent systems can exhibit measurable higher-order coordination that is not reducible to individual time series [3], and synchronization formalisms offer one route to characterising heterogeneous agent collectives [4].

The present paper differs by placing the semantic completion boundary at the centre of analysis. It neither assumes that latent attractors literally correspond to the coarse states L , G , and T , nor claims that all unnecessary continuation reflects attractor escape. Instead it proposes a measurable behavioural signature that can be tested independently and then related to hidden-state dynamics where model access permits.

10 Limitations

The framework has several limitations. Semantic completion is not uniquely defined and can be subjective for open-ended tasks. A response may contain socially useful closure that is not strictly task-essential. Behavioural classes can therefore be annotation-sensitive. The formal hazard model is a coarse-grained effective description and cannot identify a causal mechanism by itself. The conditional-independence result for multiple monitors is mathematically simple but operationally demanding because evaluator errors are often correlated. Moreover, independent termination monitors can introduce premature stopping, so any deployment must optimize both false continuation and false termination.

The use of attractor language should also be disciplined. A behavioural basin is not automatically a literal low-dimensional attractor in the transformer’s hidden state. The correct empirical sequence is behavioural detection, hidden-state measurement where possible, perturbation, and only then stronger mechanistic claims. Finally, this paper presents theoretical predictions and synthetic illustrations rather than empirical validation. The proposed experiments remain to be run.

11 Discussion

The central methodological claim is modest: semantic completion should be treated as an experimentally meaningful boundary rather than only a conversational intuition. Once the completion point is estimated independently, post-completion behaviour becomes amenable to event-history analysis. The key quantity is not simply “did the model ignore the instruction to stop?” but whether the cause-specific hazard of generic behaviour changes systematically as local semantic work disappears.

If the boundary-enrichment hypothesis is false, SBR collapses into ordinary error analysis and the framework has still provided a clear falsification route. If it is true, the completion boundary becomes a useful probe of competing behavioural tendencies. That result would support the idea that a coherent response need not correspond to one uniformly dominant regime from beginning to end.

The architectural consequence is potentially more durable than the phenomenon itself. Systems become easier to reason about when semantic generation, task verification, termination, provenance, and scope preservation are represented as distinct functions with explicit coupling. This does not require anthropomorphic “sub-minds”. It is ordinary systems engineering applied to cognitive software: separate concerns whose failure modes should not be perfectly correlated.

The longer-term implication is a shift from model-centric design to dynamics-centric design. The important unit is no longer only the capability of an individual model, but the regime entered by a network of heterogeneous processes. Stability, diversity, coordination, independent veto, and emergent behaviour then become measurable system properties. A small post-completion seam can therefore serve as an experimental seed for a much larger theory of distributed AI dynamics.

12 Conclusion

Semantic Boundary Reversion is proposed as a falsifiable post-completion phenomenon: locally unnecessary generic behaviour that appears after substantive task completion but before true semantic termination. A competing-risks formulation yields an explicit cumulative incidence and a simple boundary-enrichment criterion. The theory predicts that if local task anchoring decays after completion while generic continuation remains available, the relative hazard of reversion can increase with post-completion distance.

The framework also identifies a concrete engineering response. An independent semantic completion process increases terminal hazard and, when its errors are not perfectly correlated with the generator, reduces residual continuation risk multiplicatively. This naturally motivates heterogeneous architectures in which different components preserve different invariants rather than asking a single autoregressive process to reason, verify, govern, and terminate itself.

Whether SBR is empirically strong or weak is therefore answerable. The larger architectural question remains even if the effect is small: when a cognitive system is composed of many learned pressures, what should be centralised, what should remain independent, and which invariants must survive the moment when the original task no longer supplies the force holding the system in place?

A Illustrative competing-risk simulation

Figures 2 and 4 use synthetic rates solely to visualise the theory. Let normalised post-completion distance be $s \in [0, 1]$. The boundary-enriched illustration uses

$$\lambda_B(s) = 0.015e^{3s}, \quad \lambda_T(s) = 0.12. \quad (37)$$

The flat null uses $\lambda_B(s) = 0.06$. The independent-monitor illustration adds $\lambda_E(s) = 0.18$ to the terminal hazard. No parameters were fitted to empirical data and the figures should not be interpreted as effect-size estimates.

B Recommended reporting checklist

A study claiming SBR should report: (i) how semantic completion was defined; (ii) inter-rater agreement for τ^* ; (iii) the pre-registered behaviour taxonomy; (iv) model, decoding, and serving configuration; (v) response-length controls; (vi) cause-specific hazard estimates with uncertainty; (vii) matched null or ablation conditions; (viii) whether EOS/stopping configuration was verified; (ix) whether evaluators shared architecture with the tested model; and (x) all negative results.

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